QISKIT FALL FEST 2026 · PHASE 2 HACKATHON

Quantum Optimization Track — Judging Rubric

One 100-point scale for PS-O1, PS-O2 and PS-O3 · built from the five Phase 2 objectives and each PS's evaluation criteria · published to participants and judges

1. Eligibility gates (pass / fail — a failed gate means the submission is not ranked)

#	Gate
G1	Repository URL submitted before the end of the 24-hour round; later commits are ignored
G2	Exactly one problem statement (the team's Phase 1 PS) addressed and named in the README
G3	`main.ipynb` runs from the README and produces the required Processor B results (judges follow the README, up to two attempts, without editing code)
G4	`results/tables/results.csv` (kit format) and a technical report are present

2. Score categories (100 points)

Category	Pts	How the points are awarded
C1 Bell-state verification (Objective 1)	10	Correct $\|\Phi^+\rangle$ circuits, XX/YY/ZZ bases, adjacent and farthest pair on D, ideal and noisy, 4,096 shots (3) · correct `bell.csv`: ideal ±1 values, F = (1 + ⟨XX⟩ − ⟨YY⟩ + ⟨ZZ⟩)/4 with SE, reproducible within 2 SE (3) · interpretation of noise and distance (4). F computed as P(00)+P(11) scores 0 for correctness
C2 Processor D design (Objective 2)	20	0 if `processor_D.json` fails `validate_processor_d()`. Targets a bottleneck measured on A/B for the team's own circuit, with numbers (8) · expected effect checked against measured D results (6) · graph drawing and honest trade-off (3) · design frozen; extra noise scenarios labelled (3)
C3 Solution and benchmarking (Objective 3)	35	**Formulation 10** — O1: QUBO of the locked graph (4), brute-force C^* with both optimal partitions (3), p = 1 QAOA with ≥ 5 seeded starts (3) · O2: objective with locked μ, Σ, q (4), penalty $\lambda(\Sigma x - 2)^2$ (3), F^* over the 6 valid portfolios (3) · O3: 9-qubit QUBO, 4 slack bits, P = 5 (4), feasibility on item bits only (3), exhaustive V^* (3) **Quality 15** — objective formula, Section 3 (6) + judged (9): O1 all A/B/D ideal/noisy rows (3), mean ± spread of r over seeds (3), comparison with random and C^* (3) · O2 all rows (3), full λ sweep on D (4), comparison with random (2) · O3 all rows (3), landscape on D per spec with optimum marked (4), P_feasible vs 0.875 random (2) **Resources 10** — A/B/D table from `circuit_metrics()` with capacity rule (4), numbers reproduce (3), efficiency on D vs B at equal quality (3)
C4 Validation and reproducibility (Objective 4)	15	Principal result reproduced from a clean clone (6 first attempt / 3 second / 0) · numbers agree (4) · pinned requirements, seeds, official settings, ≤ 30 min run (3) · `results.csv` in kit format (2)
C5 Technical analysis (Objective 5)	20	O1: connectivity vs SWAPs, depth and noisy r / P_opt with own numbers (8), p or initialization effects (4) · O2: λ at which the QUBO minimum becomes valid and the feasibility–quality trade-off (8), architecture comparison (4) · O3: feasibility vs quality with the random baseline (6), landscape interpretation (4), architecture (2) · all: limitations and next steps (4), clear figures and tables (4)

Deductions: −2 per missing required run (up to −10). A changed locked instance or official setting in a required run caps C3 at half.

3. Objective formula (6 of the 15 quality points)

Uses the team's **Processor B, noisy** row (identical processor for all teams). s(x) = clip((x − x_random)/(x_reference − x_random), 0, 1); reference = organizer p = 1 QAOA on Processor B, noisy.

PS	Points	x_random → x_reference
O1	3·s(r) + 3·s(P_opt)	r 0.614 → 0.785 · P_opt 0.031 → 0.123
O2	3·s(valid frequency) + 3·t(ratio), t = clip((ratio − 0.474)/0.526, 0, 1)	valid 0.375 → 0.599 · ratio of worst valid portfolio = 0.474
O3	3·(% of optimal value / 100) + 3·s(P_feasible)	P_feasible 0.875 → 0.909

Computed with `opt_objective_points.py`; rounded to one decimal. If the reported numbers do not reproduce, these 6 points are 0.

4. Ranking, ties and procedure

- All eligible submissions are ranked together on the average of the judges' scores; the number of prize positions is set by the organizers.
- Ties (within 1.0 point): higher C3, then C2, then C4, then majority vote of the panel.
- At least two judges score each submission independently; differences above 15 points are discussed and re-scored. The panel calibrates on one submission per PS first.
- A demo or presentation only verifies the submission and adds no points. Scores, gates and one-line justifications are recorded in `score_sheet.csv`.
- Check first: O1 counts both optimal (complementary) bitstrings; O2 QUBO minimum must be invalid at λ = 0 and 0.5; O3 ~87.5% of subsets are feasible; Processor A cannot run O1 or O3 ("not executable").

5. Score sheet (one per submission, per judge — editable copy: score_sheet.csv)

Field	Max	Score	Justification (one line)
Team / repository · PS (O1/O2/O3) · judge			
Gates G1 ☐ G2 ☐ G3 ☐ G4 ☐			
C1 Bell-state verification	10		
C2 Processor D design	20		
C3 Formulation (10) · quality: formula (6) + judged (9) · resources (10)	35		
C4 Validation and reproducibility	15		
C5 Technical analysis	20		
Deductions			
Total	**100**		
B-noisy values used: O1 r, P_opt · O2 valid, ratio · O3 % optimal, P_feasible			